

28 APRILE 2026

EEG_07f × EEG_08

Dataset Build · GNN Ablation · Subject Clustering EEG-First

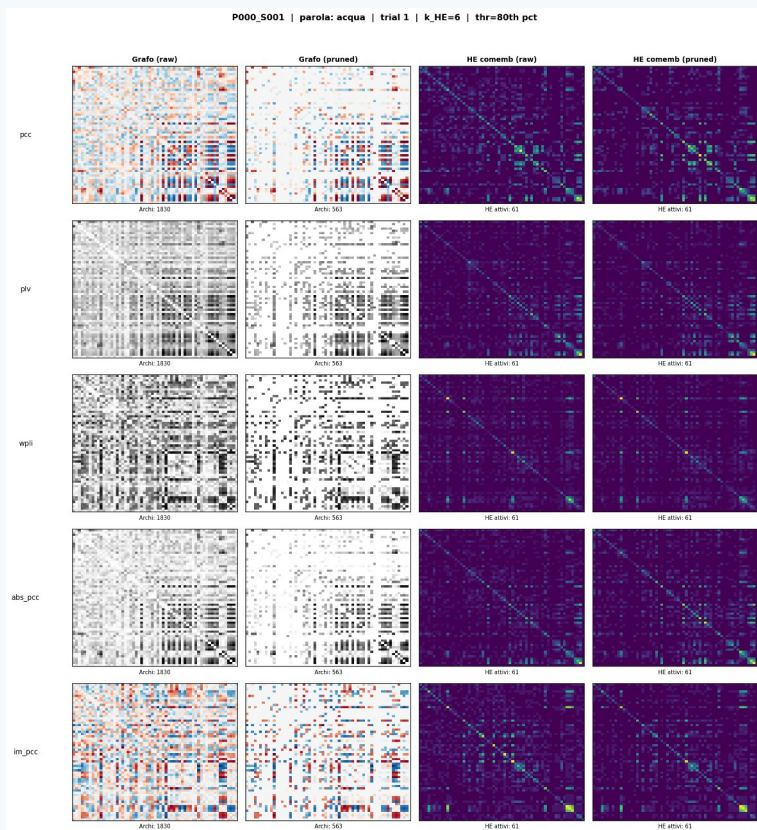
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Rel. Francesco Trovò | Decodifica Imagined Speech con Graph Neural Networks

Obiettivo: replicare DHSLP/DHSLF (Li et al. 2025) ~78% accuracy su IS con hypergraph dinamico

Graph & Hypergraph Dataset — 38.883 Trial



5 Metriche

PCC · |PCC| · Im-PCC · wPLI · PLV

2 Varianti

raw (tutte le coppie) + consensus-pruned ($\geq 2/4$ metriche)

4 Output/trial

grafo raw · grafo pruned · HG raw · HG pruned

Shape

$x = (61, 384) \cdot \text{adj} = (61, 61) \cdot H = (61, 61)$

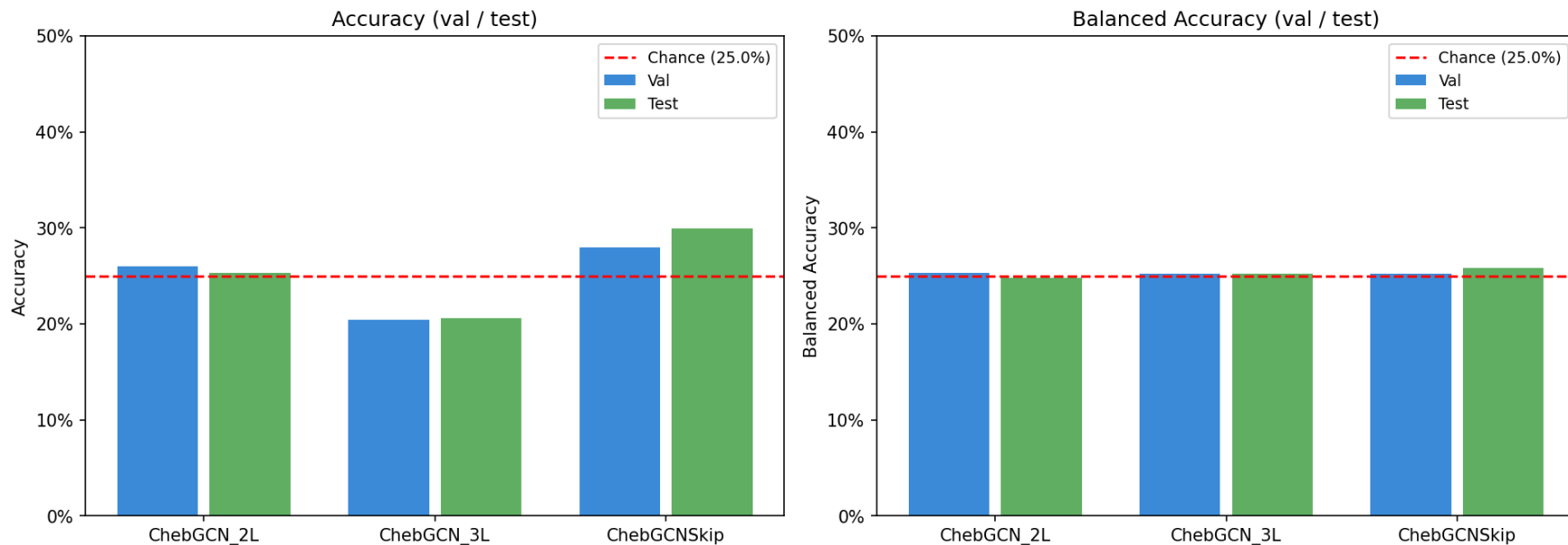
Split soggetti

TRAIN 0-49 · VAL 50-59 · TEST 60-73 (subject-indep)

1 .pt per trial — GRAPH CLASSIFICATION

GCN Spaziale: Risultati Baseline (PCC, k=6)

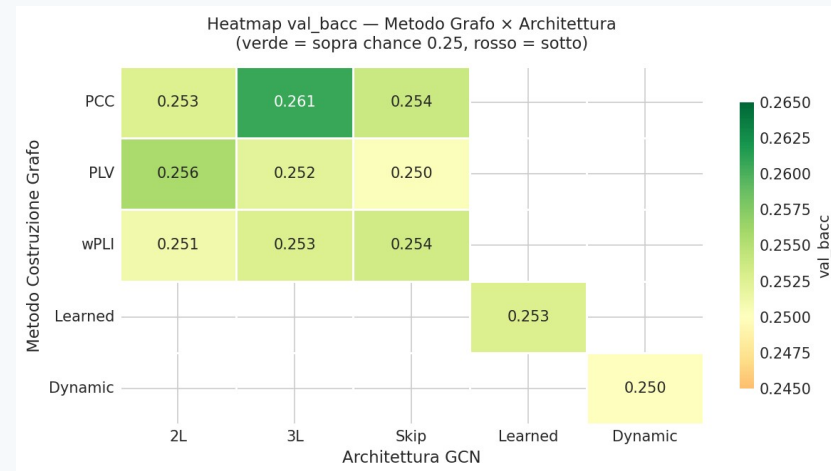
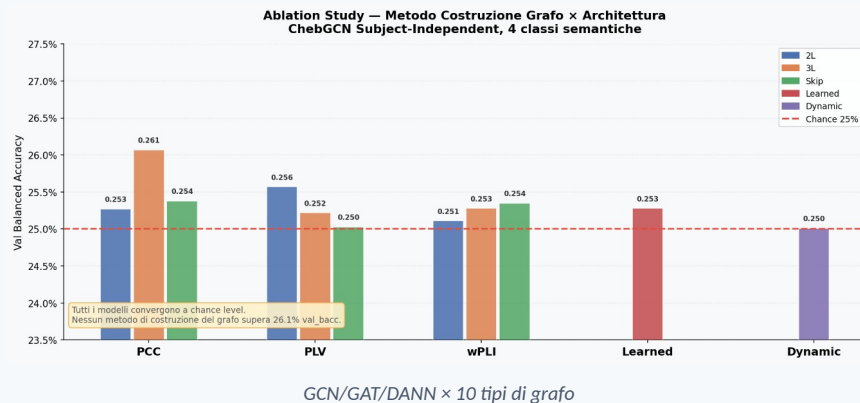
EEG_08 — GCN Spaziale (4 classi, k=6)



ChebGCN 2L/3L/Skip — val & test balanced accuracy. Chance level = 25% (4 classi concr4).

ChebGCN 2L: val 25.5% / test 24.8% • 3L: 20.5% / 20.8% (collasso) • Skip: 27.9% / 30.3% (miglior bAcc raw). Tutti al limite del chance level.

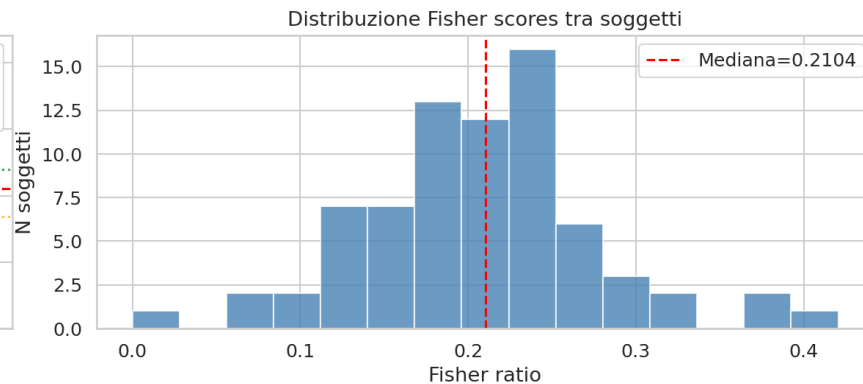
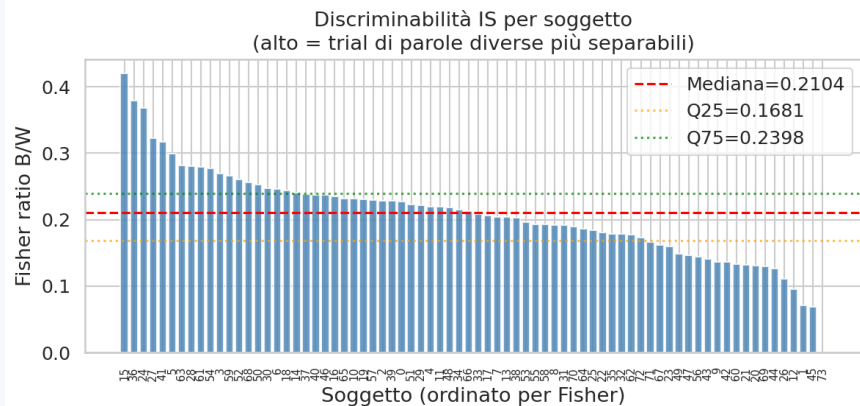
Ablation Completo: 10 Graph Types × 3 Architetture



Heatmap val_bacc — verde = sopra chance

Nessun metodo di costruzione del grafo supera il 26.1% val bAcc. Nessuna differenza significativa tra PCC/PLV/wPLI né tra GCN/GAT/DANN. Il grafo spaziale statico non è sufficiente per IS cross-subject.

Discriminabilità IS per Soggetto — Fisher B/W Ratio



Fisher ratio B/W per soggetto (74 soggetti) — ordinato per valore decrescente. Mediana = 0.21, IQR = [0.17, 0.24].

0.21

Mediana Fisher

2.5×

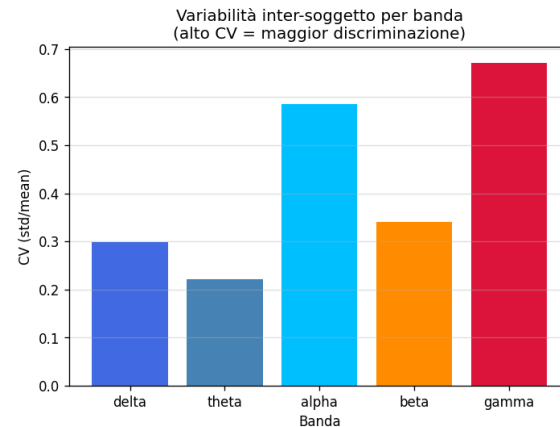
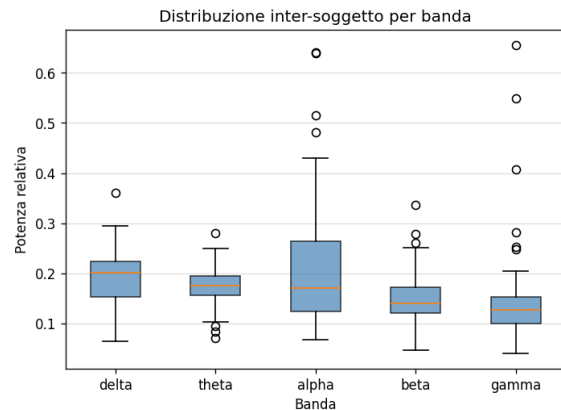
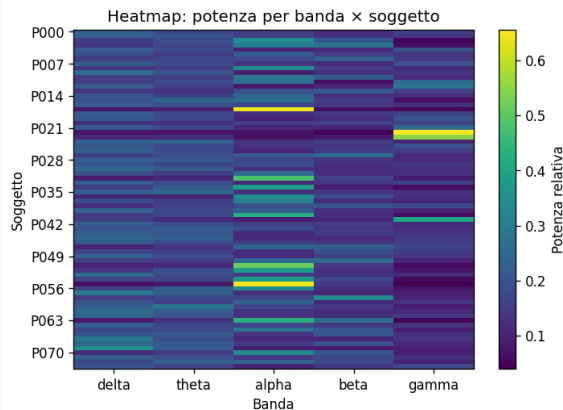
Range (min→max)

0.67

CV Gamma (max banda)

Potenza Relativa per Banda — 74 Soggetti

Potenza Relativa per Banda EEG — tutti i soggetti



Sinistra: heatmap potenza per banda × soggetto. Centro: distribuzione inter-soggetto. Destra: CV (std/mean) — gamma ha variabilità massima.

Delta — CV = 0.30

Bassa variabilità inter-soggetto

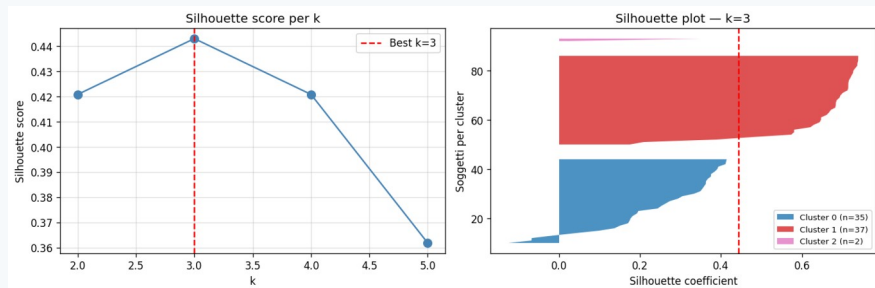
Alpha — CV = 0.59

Alta variabilità — candidato cluster

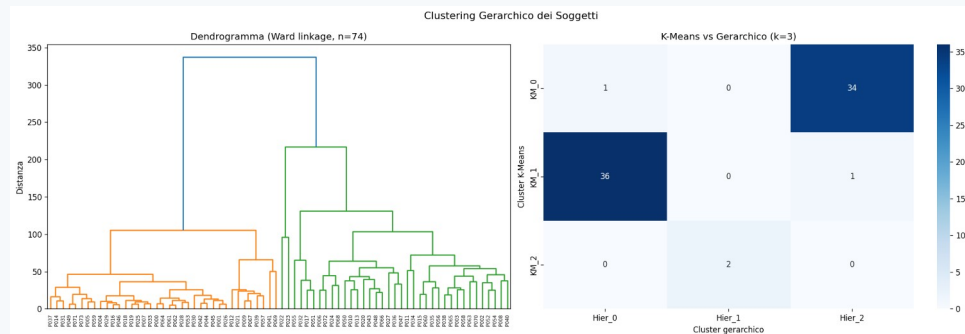
Gamma — CV = 0.67

Massima variabilità — feature top

Clustering Soggetti: k=3 Ottimale (Ward + K-Means)

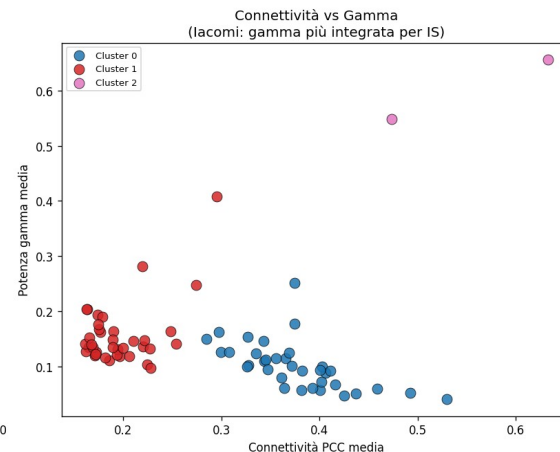
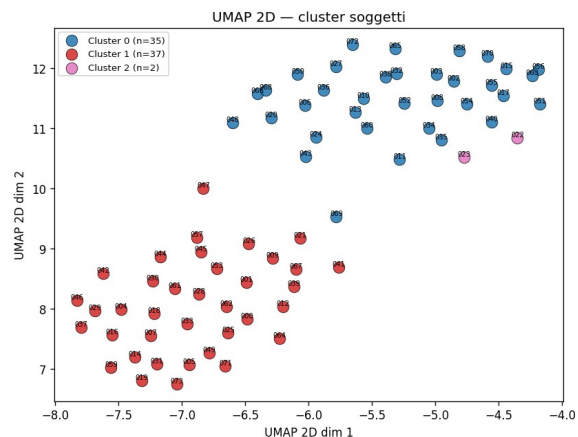
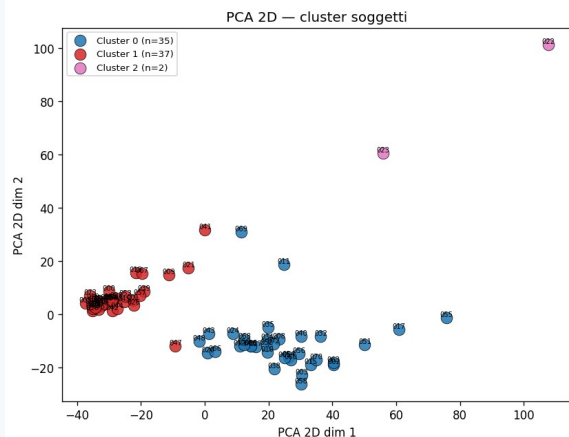


Silhouette score per k — best k=3 (0.443). Plot k=3: C0 n=35, C1 n=37, C2 n=2.

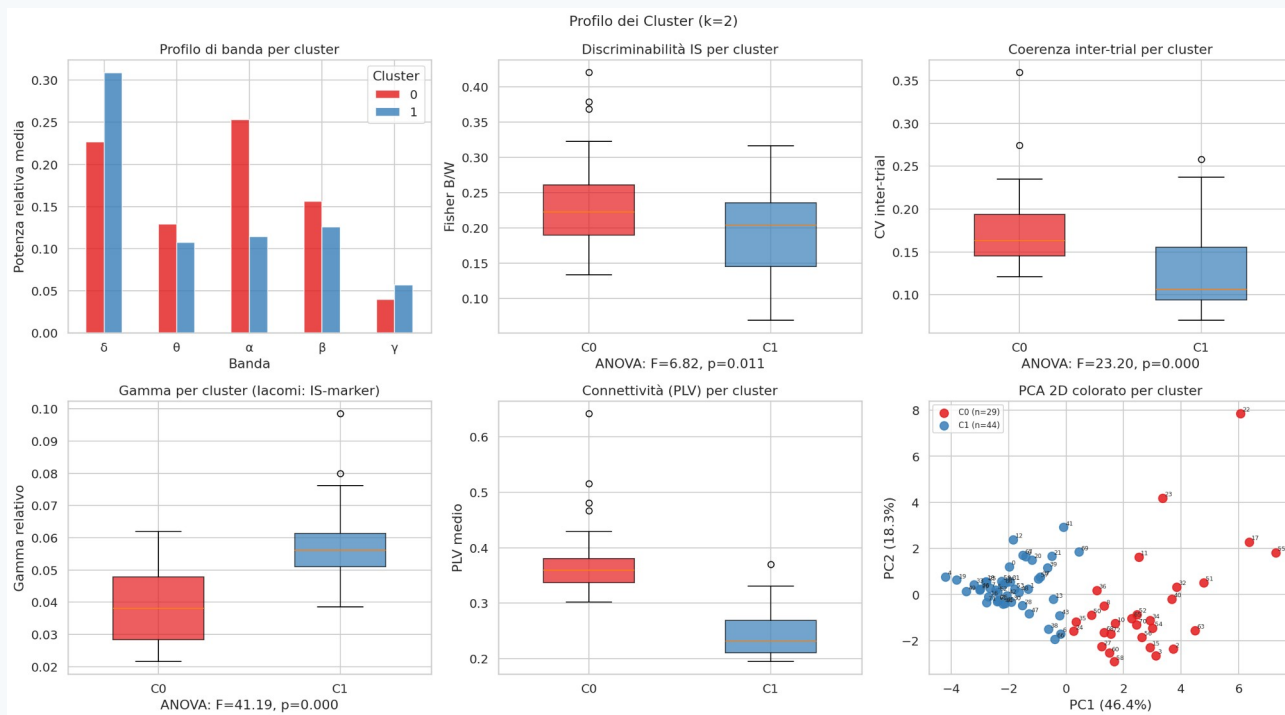


Dendrogramma Ward (n=74) — 2 macro-cluster principali. K-Means vs Gerarchico k=3: concordanza quasi perfetta.

Cluster Soggetti (K-Means k=3)



Profili dei Cluster (k=2): C0 Alta-IS vs C1 Bassa-IS



K-Means $k=2$. C0 = alta discriminabilità IS ($n=29$), C1 = bassa IS ($n=44$). ANOVA su tutte le feature.

Coer. inter-trial $F=23.20$ $p=0.000$ ***

C0 >> C1

Gamma relativo $F=41.19$ $p=0.000$ ***

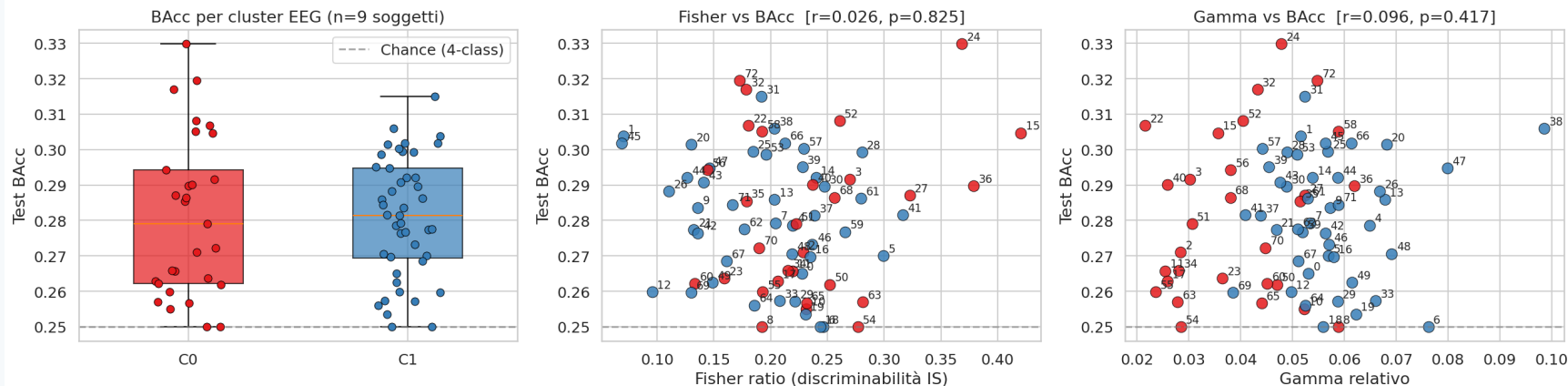
C0 < C1 (inverso!)

Fisher B/W ratio $F=6.82$ $p=0.011$ *

C0 > C1

Validazione Post-Hoc: GNN BAcc vs Cluster EEG

Validazione Post-Hoc: GCN BAcc vs Cluster EEG (n=73 soggetti)



Sinistra: GCN bAcc per cluster EEG (n=9 soggetti test). Centro: Fisher ratio vs bAcc. Destra: Gamma vs bAcc.

Risultato: Fisher vs bAcc $r=0.026$ ($p=0.825$) · Gamma vs bAcc $r=0.096$ ($p=0.417$). Il GNN non cattura la struttura EEG dei cluster soggetti. I cluster esistono nei dati ma il modello spaziale statico non li rispecchia.

Conclusioni & Prossimi Passi



EEG_07f — Dataset Pronto

38.883 trial $\times$ 5 metriche $\times$ 4 varianti (raw/pruned $\times$ grafo/ipergrafo). 1 .pt per trial. Split subject-indep confermato senza data leakage.



EEG_08 GNN — Tutti ~Chance (atteso)

GCN/GAT/DANN $\times$ 10 graph types: nessuno supera 26.1% bAcc (chance = 25%). Il grafo spaziale statico non modella la struttura temporale del segnale EEG.



EEG_08 Clustering — 2-3 Cluster Naturali

K-Means $k=3$ (silhouette 0.443). C0 alta-IS vs C1 bassa-IS: differenze significative su gamma ($F=41.19^{***}$), ITC ($F=23.20^{***}$), Fisher ($F=6.82^*$).



Post-Hoc: GNN non rispecchia cluster EEG

Fisher vs bAcc $r=0.026$, Gamma vs bAcc $r=0.096$ — nessuna correlazione. I cluster sono reali nell'EEG ma il modello spaziale non li coglie.